SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I Yazhini R (192521190) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Yazhini

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :

2.

1. CO2-AT-1

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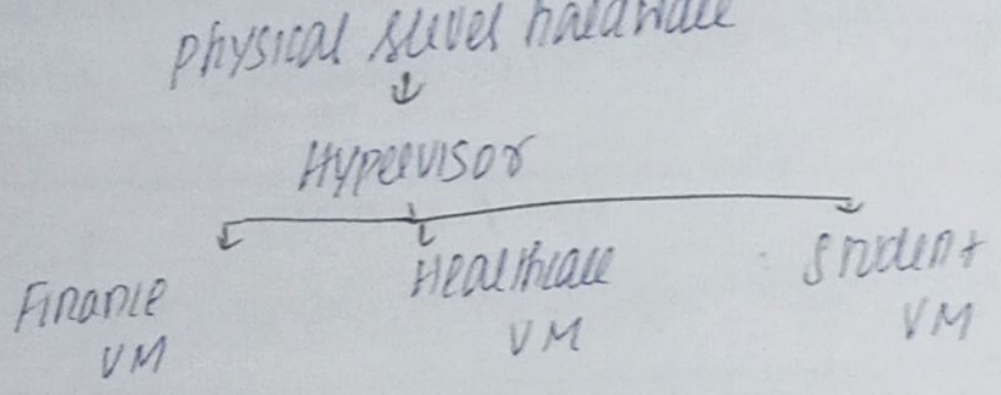
A hypervisor is software that creates and manages VMs on a physical computer. Type 1 and type 2 hypervisors differ in their architecture, performance, security, and management.

Feature	Type-1 hypervisor	Type-2 hypervisor
Architecture	Runs on physical hardware	Runs on a host OS
Performance	High performance with less overhead	More overhead due to host OS
Security	Smaller attack surface; strong isolation when properly configured	Depends on the security of host OS & hypervisor
manageability	Supports centralized enterprise management	Generally based for individual users to manage
Resource usage	Efficient use of CPU and memory	Uses resources for both the host OS and VMs
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation
Applications	Cloud datacenters and Enterprise servers	Testing, development and personal use

Type-1 hypervisors provide better performance because they run directly on hardware without depending a general-purpose host OS

Type-2 hypervisors are useful for testing and development but introduce additional overhead and depend on the host OS.

2. Virtualization allows finance, healthcare, and student portals to run in separate VM on the same physical server.



How virtualization provides Isolation:
The hypervisor allocates separate CPU, memory, storage and network resources to each VM. This prevents direct access between workloads. Network segmentation and access controls provide additional security.

Two major attack surfaces:
1. Hypervisor vulnerabilities:
Attackers may exploit hypervisor weakness to access other VMs. Regular patching and secure configuration reduce this risk.

2. Virtual network attacks:
Misconfigured virtual networks may allow unauthorized access. Firewalls, network segmentation, and encryption help protect communication.

Virtualization enables secure workload separation on shared hardware when combined with proper hypervisor and network security.

COURSE

The VM host shows high resource usage, which can reduce performance and increase startup time.

Metric	Value	Likely problem
CPU	92%	CPU resource contention
Memory	88%	Low available memory
Storage I/O latency	35ms	Slow disk operations
VM boot time	170s	Delayed startup

LIKELY BOTTLENECKS:

High CPU and memory usage indicate resource shortages. Storage latency of 35ms suggests slow disk access, which may also delay VM startup.

CORRECTIVE ACTIONS:

Add CPU and memory resources where necessary, redistribute VM across hosts, optimize storage, and stagger VM startups. Continuous monitoring helps identify bottlenecks and improve performance.

CONCLUSION:

Resource balancing and storage optimization can reduce boot time and improve overall VM performance.

4. Software Defined Datacenter (SDDC)

An SDDC uses software to manage and automate compute, storage, and networking resources. Unlike traditional datacenters, it reduces manual configuration and enables faster resource allocation.

Feature	Traditional datacenter	SDDC
Compute	physical server management	virtual machines
Storage	separate storage devices	pooled virtual storage
Network	manual configuration	software-controlled network
Provisioning	slow and manual	fast and automated
Scalability	more manual labor	dynamic resource allocation
Management	separate hardware tools	centralized software management

Computer virtualization:
Runs multiple VM on physical servers, improving resource utilization and deployment speed

Storage virtualization:
Combines storage devices into logical pools, making storage allocation flexible and easier to manage

Network virtualization:
Creates software-based networks, enabling centralized configuration, traffic control, and workload isolation

SDDC improves agility through automation, faster deployment, easy scaling and better resource utilization.

5 Multi-cloud Federation and Identity Mismatches
Multi-cloud Federation enables users to access services across different cloud providers through a trusted identity system. Identity mismatches occur when providers use different user IDs, authentication methods, and permission roles, causing login failures or incorrect access

User

↓

Central Identity Provider (IdP)

↓

SAML / OpenID Connect

↓

Cloud A Cloud B

proposed Identity design:

Use a central Identity provider to authenticate users across cloud platforms. SAML or OpenID can be enables secure Identity Exchange. While role mapping ensures correct permissions

Privacy and security:

Share only necessary user information, encrypt data during transmission, and use short-lived tokens. Audit logs and regular permission reviews help detect suspicious activities and protect personal data

conclusion:

Centralized Identity Management, Federation standards, and correct role mapping resolve Identity mismatches while maintaining security and privacy across multiple clouds